INCH-POUND

MIL-DTL-83725/10B
17 February 2022
SUPERSEDING
MIL-R-83725/10A(USAF)
3 October 1986

DETAIL SPECIFICATION SHEET

RELAYS, VACUUM, SPST, N/C, 12 AMPERES DC OR 60 Hz RMS, 10 KILOVOLTS DC OR 60 Hz PEAK

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-DTL-83725.

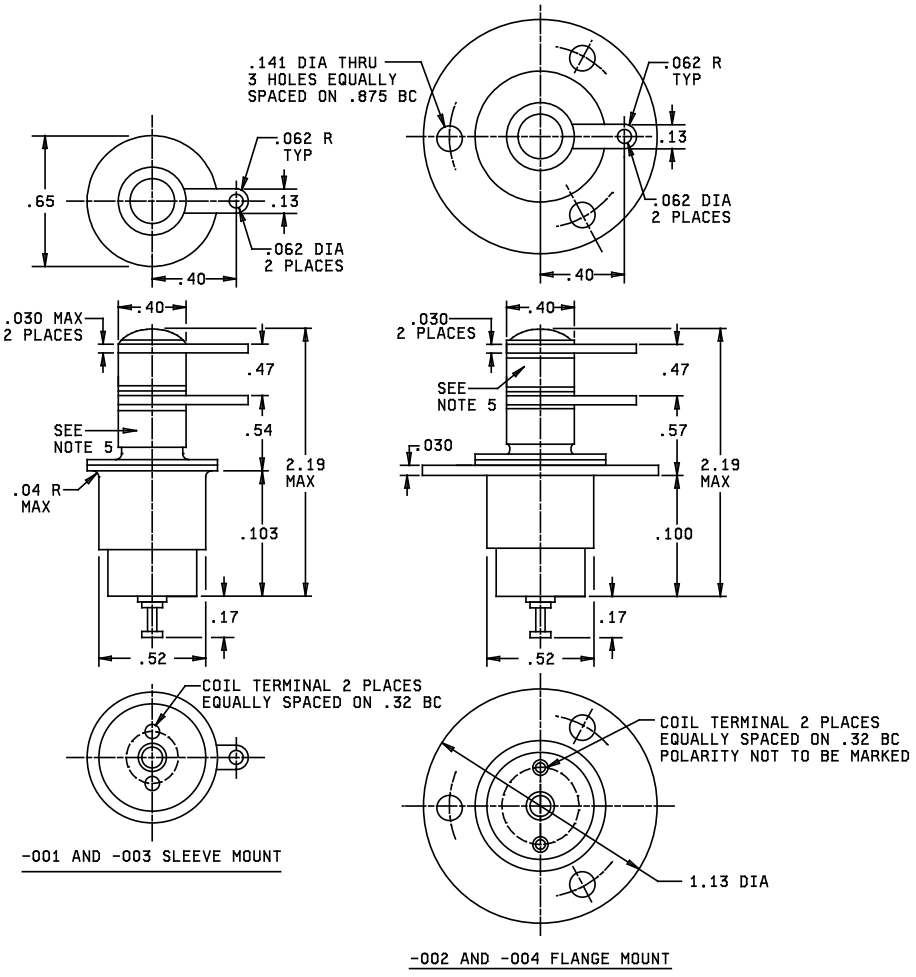
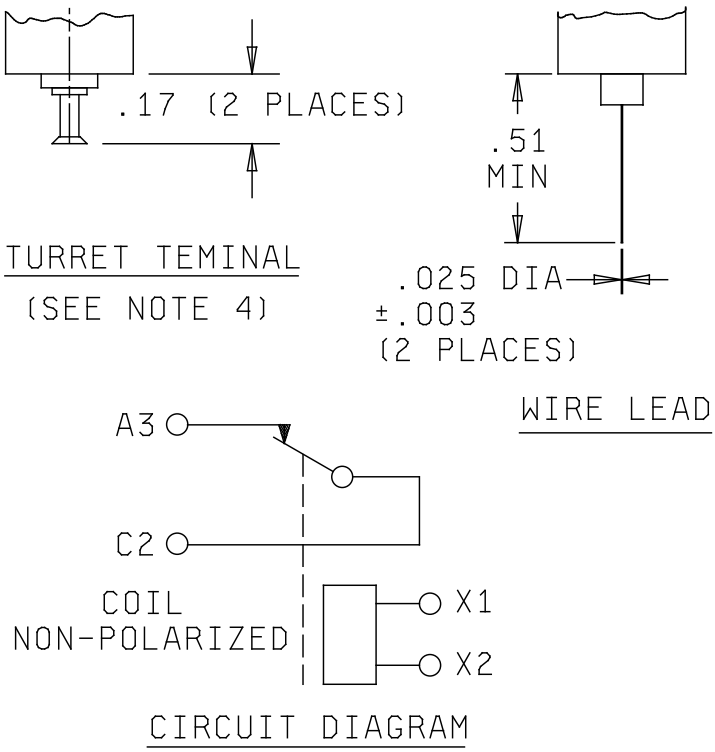


FIGURE 1. Relay, dimensions and configurations.

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Inches	mm	Inches	mm
.015	0.38	0.51	12.95
.025	0.64	0.52	13.21
.03	0.76	0.54	13.72
.04	1.02	0.57	14.48
.062	1.57	0.65	16.51
.13	3.30	0.875	22.23
.141	3.58	1.00	25.4
.17	4.32	1.03	26.16
.32	8.13	1.13	28.70
.40	0.16	2.19	55.63
.47	1.94		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three-place decimals and ±.03 (0.76 mm) for two-place decimals.
4. Shape of turret terminals optional, however they shall accommodate two number 22 AWG wires as a minimum.
5. Glazed finish on ceramic insulators is optional.
6. Sleeve mount relays are mounted by 0.52 inch (13.21 mm) diameter of body of coil housing or by 0.65 inch (16.51 mm) diameter lip in center of relay.

FIGURE 1. Relay, dimensions and configurations - Continued.

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REQUIREMENTS:

CONTACT DATA:

Configuration: SPST, N/C, ground isolated.

Arrangement: 1 form B.

Load ratings:

Resistive: (Carry only):

12 amperes dc or 60 Hz rms.

10 amperes rms, 2.5 MHz.

6 amperes rms, 16 MHz.

Rated operating voltage:

10 kilovolts peak, 60 Hz or dc.

7 kilovolts peak, 2.5 MHz.

6 kilovolts peak, 16 MHz.

Contact resistance:

Rated Life:

Before: .020 ohm, maximum.

During: .100 ohm, maximum.

After: .100 ohm, maximum.

Capacitance:

1.6 picofarads, maximum across open contacts.

1.6 picofarads, maximum between open contacts and ground.

COIL DATA: (See table I).

Duty rating: Continuous.

Operate time: 15 milliseconds, maximum over temperature range with nominal coil voltage.

Release time: 15 milliseconds, maximum over temperature range from nominal coil voltage.

ELECTRICAL DATA:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and housing at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

At atmospheric pressure: 60 Hz.

Between all mated contacts in the open position: 12 kV peak.

Between high voltage terminals and housing: 12 kV peak.

Between coil and housing: 500 V rms, 60 Hz.

Ground isolated: Ground isolated relays have operating elements insulated from the ground plane to provide absolute voltage isolation between the frame and the high voltage contacts.

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ENVIRONMENTAL DATA:

Temperature range: -55°C to +125°C.

Vibration: MIL-STD-202- 204, test condition C (10 to 55 Hz, .06 inch DA/55 to 2000 Hz, 10 g).

PHYSICAL:

Terminal strength: 5 pounds pull.

Dimensions and configuration: See figure 1.

Terminations: See figure 1.

Weight: 1 ounce, maximum.

LIFE TEST REQUIREMENTS:

Mechanical cycling: 1,000,000 cycles. 2 samples units (cycling rate, 36,000 per hour maximum; contact current shall not exceed 10 milliamperes).

QUALITY ASSURANCE:

Dielectric withstanding voltage:

Tests to be conducted at atmospheric pressure rating only.

Duration of application: 5 to 10 seconds at a 10 percent increase in the dielectric withstanding voltage.

PART NUMBER: See table I.

TABLE I. Dash numbers and characteristics.

Dash number M83725/10 -	Coil data (V dc)		At 25°C			Over temperature range				Mounting	Terminal type
	Nominal 1/	Maximum	Coil resistance ohms ±10%	Pick up voltage (V dc)	Hold voltage (V dc)	Drop-out Voltage (V dc)	Pick-up Voltage (V dc)	Hold Voltage (V dc)	Drop-out Voltage (V dc)		
-001	26.5	32	290	16	10	1.0	22	14	0.7	Sleeve	Turret
-002	"	"	"	"	"	"	"	"	"	Flange	Turret
-003	"	"	"	"	"	"	"	"	"	Sleeve	Wire
-004	"	"	"	"	"	"	"	"	"	Flange	Wire
-005	12	16	70	8.0	5.0	0.5	11	7.0	0.3	Sleeve	Turret
-006	"	"	"	"	"	"	"	"	"	Flange	Turret
-007	"	"	"	"	"	"	"	"	"	Sleeve	Wire
-008	"	"	"	"	"	"	"	"	"	Flange	Wire

1/ CAUTION: Use of any coil voltage less than the nominal coil voltage will compromise the operation of the relay.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Referenced documents. In addition to MIL-DTL-83725, this document references the following:

MIL-STD-202-204

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA - CC

Preparing activity:
DLA-CC

Review activity:
Navy - OS

(Project 5945-2022-013)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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